

VECTORS	
Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2}$
	$\ v\ = \sqrt{v^T v}$
	$\nabla \ v\ = \frac{1}{\ v\ } v^T$
Distance	$d(u, v) = \ u - v\ $
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$

MATRICES	
Determinant:	
$\det(a) = a$	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$
$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$	
Inverting:	
$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$	
Transposing:	
$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^T)^T = A \quad (A + B)^T = A^T + B^T$	
$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5) \quad (\lambda A)^T = \lambda A^T \quad (AB)^T = B^T A^T$	
Matrix multiplication:	

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^T)^T = A \quad (A + B)^T = A^T + B^T$$

$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5) \quad (\lambda A)^T = \lambda A^T \quad (AB)^T = B^T A^T$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$

$$= \begin{pmatrix} (2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8) & (2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9) \\ (4 \cdot 6 + -1 \cdot 1 + 7 \cdot 8) & (4 \cdot 4 + -1 \cdot 0 + 7 \cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

Affine transformation:

$$M \in \mathbb{R}^{m \times n}, \quad A_{b,M} : \begin{cases} \mathbb{R}^m \rightarrow \mathbb{R}^n \\ x \mapsto b + M \cdot x \end{cases}$$

INDICES

$$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, A \cdot B \in \mathbb{R}^{n \times r}$$

$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$$

$$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ki}$$

$$a^T \cdot B \cdot b = \sum_{ij} (a^T)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$$

$$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$$

$$(E_{rat})_{ijk} = \delta_{ri} \delta_{aj} \delta_{bk}$$

BASIS VECTORS

$$\mathbb{R}^3$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$

$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

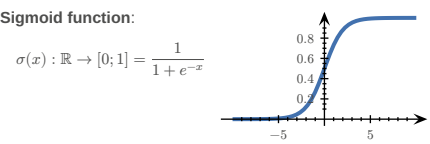
AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M \cdot x$$

DERIVATIVES	
$\frac{1}{x} \rightarrow -\frac{1}{x^2}$	$x \rightarrow 1$
$e^x \rightarrow e^x$	$x^n \rightarrow nx^{n-1}$
$\ln(x) \rightarrow \frac{1}{x}$ for $x > 0$	$\sqrt{x} \rightarrow \frac{1}{2\sqrt{x}}$
$a^x \rightarrow \ln(a) \cdot a^x$ for $a > 0, a \neq 1$	$e^{-x} \rightarrow -e^{-x}$
$\sin(x) \rightarrow \cos(x)$	$\ln(y \cdot x) \rightarrow \frac{1}{y} \text{ for } x > 0$
$\cos(x) \rightarrow -\sin(x)$	$\log_a(x) \rightarrow \frac{1}{\ln(b) \cdot x}$
$\tan(x) \rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\sin(2x) \rightarrow 2 \cos(2x)$
$\cot(x) \rightarrow -\frac{1}{\sin^2(x)}$	$\cos(ax) \rightarrow -a \sin(ax)$
$\arccos(x) \rightarrow -\frac{1}{\sqrt{1-x^2}}$	$\operatorname{arccot}(x) \rightarrow -\frac{1}{1+x^2}$
$\arcsin(x) \rightarrow \frac{1}{\sqrt{1-x^2}}$	$\operatorname{arcsin}(x) \rightarrow \frac{1}{\sqrt{1-x^2}}$
$\arctan(x) \rightarrow \frac{1}{1+x^2}$	$\operatorname{arctan}(x) \rightarrow \frac{1}{1+x^2}$

Term	Definition
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha \cdot f(x))' = \alpha \cdot f'(x)$
Product rule	$(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES	
Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
Softmax function	TODO:
RSS	$RSS = \sum_i (y_i - f(x_i))^2$
n -dimensional curve	$c: \mathbb{R} \rightarrow \mathbb{R}^n$
n -dimensional hyper-surface	$s: \mathbb{R}^n \rightarrow \mathbb{R}$
Reparametrization	$c(t) \rightarrow d(s) = c(h(s))$



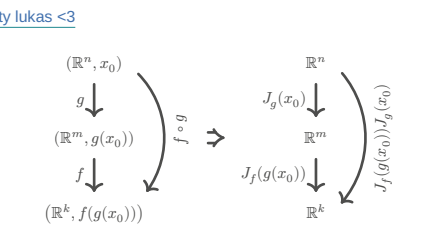
DERIVATIVES	
Function	Derivative
$f: \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \frac{d}{dx} f_2(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} \\ x \mapsto y \end{cases} = \nabla f = \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right)$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^m \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} \\ x \mapsto y \end{cases} = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0} = \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE	
TODO: example FS2024	
$f(x, y) = x^2 + 2y$	
$\frac{\partial f}{\partial x} = 2x$	

GRADIENTS	
Common gradients:	
$f(x) = \ x\ ^2$	$\Rightarrow \nabla f = 2x$
$f(x) = a^T x + c$	$\Rightarrow \nabla f = a$
$f(x) = x^T A x$	$\Rightarrow \nabla f = (A + A^T)x$
$f(x) = \frac{1}{2} x^T A x + b^T x + c$	$\Rightarrow \nabla f = \frac{A + A^T}{2} x + b$
	$= Ax + b$ if $A^T = A$
$f(x) = g(Ax + b)$	$\Rightarrow \nabla f = A^T \nabla g(Ax + b)$
$g: \mathbb{R}^m \rightarrow \mathbb{R}$	
JACOBIAN MATRIX	
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$x_0 \in \mathbb{R}^n$
$J_f(x_0): \mathbb{R}^{m \times n}$	$J_f(x_0)_{ij} = \frac{\partial}{\partial x_j} f_i(x_0)$
$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_1} & \frac{\partial x_1}{\partial y_1} \\ \frac{\partial x_2}{\partial x_1} & \frac{\partial x_2}{\partial y_1} \end{pmatrix}$

$$J_f(x_0): \mathbb{R}^{m \times n} \quad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j} f_i(x_0)$$

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_1} & \frac{\partial x_1}{\partial y_1} \\ \frac{\partial x_2}{\partial x_1} & \frac{\partial x_2}{\partial y_1} \end{pmatrix}$$



Linearisation:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x_0 \in \mathbb{R}^n$$

$$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$$

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad g: \mathbb{R}^m \rightarrow \mathbb{R}^k, \quad h: \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$$

$$J_h = J_g(f(x)) \cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M \cdot x \Rightarrow J_f = M$$

$$f(x) = x \Rightarrow J_f = \mathbb{1}$$

$$f(x) = \|x\|^2 \Rightarrow J_f = (2x)^T$$

TODO:
 J_s for softmax $s_i = e^{x_i} / \sum_j e^{x_j} ; J_{ij} = s_i(\delta_{ij} - s_j)$

HYPER-SURFACES

Chain rule:

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad g: \mathbb{R}^m \rightarrow \mathbb{R}, \quad h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$$

$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f) |_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

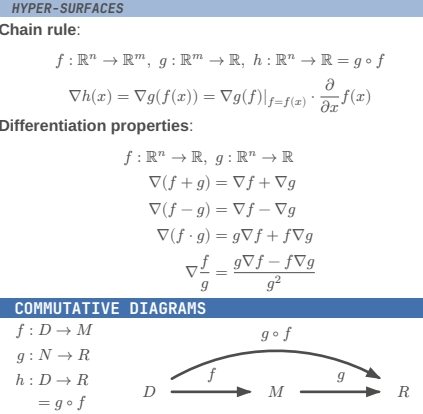
$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad g: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\nabla(f + g) = \nabla f + \nabla g$$

$$\nabla(f - g) = \nabla f - \nabla g$$

$$\nabla(f \cdot g) = g \nabla f + f \nabla g$$

$$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$$



GEOMETRY OF HYPER-SURFACES	
$f: \mathbb{R}^2 \rightarrow \mathbb{R}$	
Tangent space: $\alpha, \beta \in \mathbb{R}, T =$	
$\left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$	
Graph of a linearisation:	
$l(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) = T_{p_0}$	

SECOND DERIVATIVE

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (f(x, y)) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$$

HESSIAN MATRIX

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n \times n}$$

$$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$

$$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$$

$$\partial_i \partial_j f = \partial_j \partial_i f$$

Common Hessians:

$$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$$

$$f(x) = g(Ax + b) \Rightarrow H_f = A^T H_g(Ax + b) A$$

$$g: \mathbb{R}^m \rightarrow \mathbb{R}$$

EXTREMAL POINTS

Let $z = f(c(t))$, c a curve passing the stationary point $c_0 = (x_0, y_0)$ at $t = t_0$ and

$$\frac{d^2 z}{dt^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T} \cdot \underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0) \end{pmatrix}}_H \cdot \underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0) \end{pmatrix}}_v$$

1) If $\forall v \in \mathbb{R}^2 \setminus \{0\} (v^T H v > 0)$, then $d^2 z / dt^2 > 0$ defines a **local minimum**

2) If $\forall v \in \mathbb{R}^2 \setminus \{0\} (v^T H v < 0)$, then $d^2 z / dt^2 < 0$ defines a **local maximum**

3) If $\forall v \in \mathbb{R}^2 \setminus \{0\} (v^T H v < 0 \vee v^T H v > 0)$, the point $c_0 = (x_0, y_0)$ defines a **saddle point**

TODO:

$$(J_u)_{ij} = \frac{\partial u_i}{\partial x_j}, (H_g)_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$$

$$\partial_j (u_i v_j) = (\partial_j u_i) v_j + u_i (\partial_j v_j)$$

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$= (v_1, v_2, v_3) \begin{pmatrix} v_1 h_{11} + v_2 h_{12} + v_3 h_{13} \\ v_1 h_{21} + v_2 h_{22} + v_3 h_{23} \\ v_1 h_{31} + v_2 h_{32} + v_3 h_{33} \end{pmatrix}$$

$$= v_1^2 h_{11} + v_1 v_2 h_{12} + v_1 v_3 h_{13} + v_1 v_2 h_{21} + v_2^2 h_{22} + v_2 v_3 h_{23} + v_1 v_3 h_{31} + v_2 v_3 h_{32} + v_3^2 h_{33}$$

$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$$

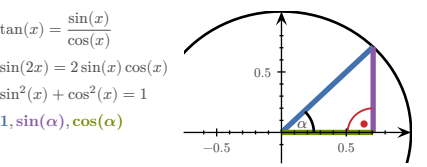
Technique of completing squares

$$\frac{x^2}{\sqrt{}} - \frac{4xy}{\div 2x} + 1 = (x - 2y)^2 - (-2y)^2 + 1 = (x - 2y)^2 - 4y^2 + 1$$

Quadratic form

$$M \in \mathbb{R}^{n \times n}, \quad q_M(v) = v^T M v$$

MISC	
$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
TRIGONOMETRY	
x	0
$\sin(x)$	0
$\cos(x)$	1
$\tan(x)$	0
$30^\circ = \pi/6$	$45^\circ = \pi/4$
$60^\circ = \pi/3$	$90^\circ = \pi/2$
$\sqrt{3}/2$	$\sqrt{3}/2$
$1/2$	$1/2$
$1/\sqrt{3}$	$\sqrt{3}$



GRADIENT DESCENT / NEWTON'S METHOD	
GD	$x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$
N	$x_{i+1} = x_i - \gamma \cdot (H_f(x_i))^{-1} \cdot \nabla f(x_i)$
γ = step size	
termination criterion $\varepsilon > x_{i+1} - x_i $	

PROBABILITY THEORY	
Term	Definition
~	Uniformly distributed
CDF	Cumulative Distribution Function

KOLMOGOROV AXIOMS

$$\mathbb{P}(\Omega) = 1$$

$$\mathbb{P}(\emptyset) = 0$$

$$\forall A, B \subset \Omega, \quad A \cap B = \emptyset \Rightarrow (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$$

$$\forall A, B \subset \Omega \quad (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$$

DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega = \{e_1, e_2, \dots, e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, \dots, n\}, \quad E \subset \Omega$$

$$\mathbb{P}(E) = \sum_{e \in E} p(e)$$

$$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$$

RULES

$$\forall e \in \Omega (0 \leq p(e) \leq 1)$$

$$\sum_{e \in \Omega} p(e) = 1$$

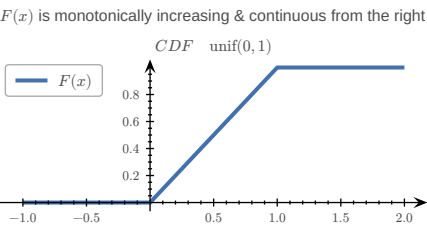
$$f(t) \geq 0 \text{ for } t \in \mathbb{R}$$

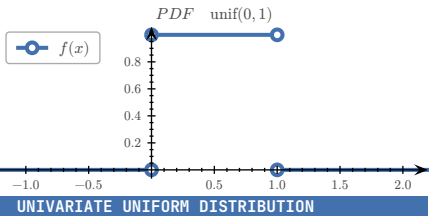
$$\int_{-\infty}^{\infty} f(t) dt = 1$$

$$0 \leq F(x) \leq 1$$

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow \infty} F(x) = 1$$





$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$

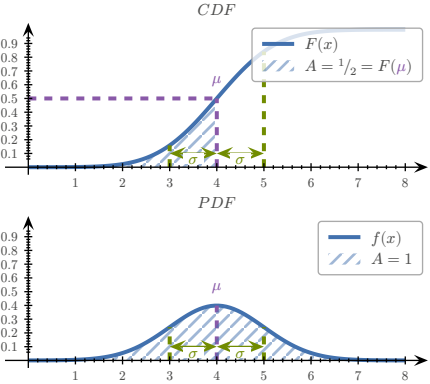
$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a; b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) = F'_X(x)$$
$$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) = \mathbb{1}_{[a;b)}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b;\infty)}(x)$$
$$= \int_{-\infty}^x f(t) \, dt$$
$$\mathbb{P}(X \leq x) = \int_{-\infty}^x \frac{\mathbb{1}_{(a;b)}(t)}{b-a} \, dt \stackrel{x \in (a;b)}{=} \frac{x-a}{b-a}$$
$$\mathbb{P}(E) = \int_{x \in E} f(x) \, dx$$

UNIVARIATE NORMAL DISTRIBUTION

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

mean value (μ), standard deviation (σ)

TODO:



Standard normal distribution: $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-1/2 x^2}$
General normal distribution: $f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$
Error function: $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \, dt$

MULTIVARIATE NORMAL DISTRIBUTION

TODO:

$\Sigma = \text{covariance matrix}, \quad V : \Omega \rightarrow \mathbb{R}^n$

$$V \sim \mathcal{N}(\mu, \Sigma) \text{ if } PDF_V = f_V(v) = \frac{e^{-\frac{1}{2}(v-\mu)^T \Sigma^{-1}(v-\mu)}}{\sqrt{(2\pi)^n \det \Sigma}}$$
$$q_{\Sigma^{-1}}(v-\mu) = -\frac{1}{2}(v-\mu)^T \Sigma^{-1}(v-\mu)$$

Sample mean: $\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i$
Sample covar. matrix: $Q = \frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})(v_i - \bar{v})^T$

EXAMPLES

TODO:

WORKING WITH INDICES

$x, y \in \mathbb{R}^n, \quad f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(x) = (x-y)^T(3x-y) = \sum_{i=1}^n (x_i - y_i)(3x_i - y_i)$$
$$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$$
$$f(x) = \ln(x_1 \cdot x_2 \cdot \dots \cdot x_n) - |x|^2 = \sum_{i=1}^n (\ln(x_i) - x_i^2)$$
$$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$$